Summary

Domains & Setup

Atari Domains:

- Qbert
- Donkey Kong

Algorithms Used:

- Deep Q Network (DQN)
- Proximal Policy Optimization (PPO)

Hyperparameter Variations:
Each game was trained with specified algorithm and parameters, tested separately with different learning rates.

Domain	Algorithm	Parameters	Learning Rate(s)
Qbert	DQN	buffer=100K, batch=32, train_freq=4, target_update=1000	1e-4, 5e-5
Qbert	PPO	n_steps=128, batch=256, n_epochs=4, ent_coef=0.01, clip_range=0.1, n_envs=8	3e-4, 1e-4
Donkey Kong	DQN	same as Qbert (DQN)	1e-4
Donkey Kong	PPO	same as Qbert (PPO)	3e-4

Saved Checkpoints:

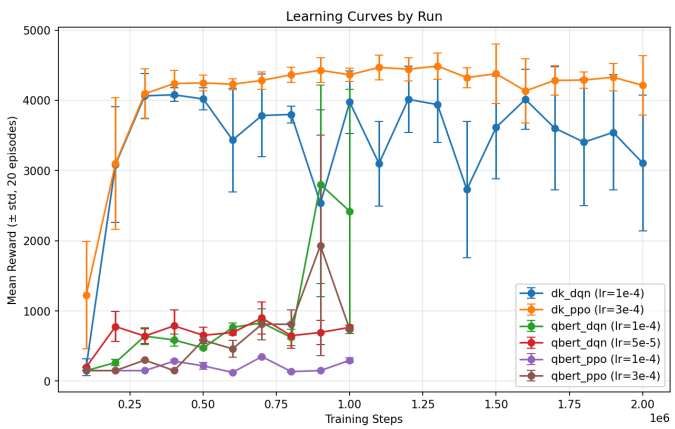
- Qbert: Checkpoint saved every 100,000 timesteps (1,000,000 total)
- Donkey Kong: Checkpoint saved every 100,000 timesteps (2,000,000 total)

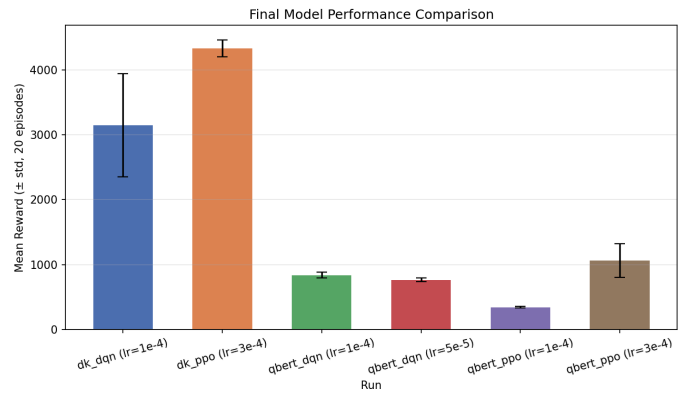
Evaluation Scores

Final Model Performance

Domain	Algo	Learning Rate	Mean Reward	Std	Min	Max
Qbert	DQN	1e-4	838.8	±47.7	800	950
Qbert	DQN	5e-5	762.5	±29.0	725	800
Qbert	PPO	3e-4	1062.5	±259.0	700	1275
Qbert	PPO	1e-4	342.5	±11.5	325	350
Donkey Kong	DQN	1e-4	3145.0	±797.2	2000	4100
Donkey Kong	PPO	3e-4	4330.0	±130.8	4100	4600

Learning Curves and Performance Comparison





Short Observations

- PPO outperformed DQN on both domains
- Lower learning rates led to worse performance, and higher learning rates led to better performance
- Donkey Kong agent learned to prioritize jumping over barrels over completing levels, leading to the learning curve stabilizing quickly
- Qbert was slower to progress, but appeared to make drastic improvements towards the end